I SEMESTER	L	T	P	C
	-	-	3	1.5
20CS1L01 : C PROGRAMMING LAB				

COURSE OBJECTIVES:

1. To impart knowledge on various Editors, Raptor.
2. To make the students understand the concepts of C programming.
3. To nurture the students on Control Structures and develop different operations on arrays
4. To make use of String fundamentals and modular programming constructs.
5. To implement programs using dynamic memory allocation.
6. To explain the concepts of Structure, Unions and files for solving various problems.

COURSE OUTCOMES:

CO1: Implement basic programs in C and design flowcharts in Raptor.
CO2: Use Conditional and Iterative statements to solve real time scenarios in C.
CO3: Implement the concept of Arrays and Modularity and Strings.
CO4: Apply the Dynamic Memory Allocation functions using pointers.
CO5: Develop programs using structures, and Files.

LIST OF EXPERIMENTS:

1. Introduction to Algorithms and Flowcharts

1.1) Implement Algorithm Development for Exchange the values of Two numbers.

1.2) Given a set of n student's examination marks (in the range 0-100) make a count of the number of students that passed the examination. A Pass is awarded for all of 50 and above.

1.3) Given a set of n numbers design an algorithm that adds these numbers and returns the resultant sum. Assume N is greater than or equal to zero.

2. Introduction to C Programming

2.1) Exposure to Turbo C, Code Blocks IDE, Dev C++, Falcon C++.
2.2) Writing simple programs using printf(), scanf() .

3. Raptor

3.1) Introduction to Raptor.
3.2) Draw a flow chart to find the Sum of 2 numbers.
3.3) Draw a flow chart to find Simple interest.

4. Basic Math

4.1) Write a C Program to convert Celsius to Fahrenheit and vice versa.
4.2) Write a C Program to find largest of three numbers using ternary operator.
4.3) Write a C Program to Calculate area of a Triangle using Heron's formula.

5. Control Flow- I

5.1) Write a C Program to Find Whether the Given Year is a Leap Year or not.
5.2) Write a C program to find the roots of a Quadratic Equation.
5.3) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using Switch…case.

6. Control Flow- II

6.1) Write a C Program to Find Whether the Given Number is Prime number or not.
6.2) Write a C Program to Find Whether the Given Number is Armstrong Number or not.
6.3) Write a C program to print Floyd Triangle.

7. Control Flow- III

7.1) Write a C program to find the sum of individual digits of a positive integer.

7.2) Write a C program to check whether given number is palindrome or not.

7.3) Write a C program to read two numbers, x and n, and then compute the sum of the geometric progression $1+x+x^2+x^3+\ldots\ldots\ldots\ldots+x^n$.

Practice Programs:

1. Write a C program to print all natural numbers from 1 to n. - using while loop
2. Write a C program to print all natural numbers in reverse (from n to 1). - using while loop
3. Write a C program to print all alphabets from a to z. - using while loop
4. Write a C program to print all even numbers between 1 to 100. - using while loop
5. Write a C program to print sum of all even numbers between 1 to n.
6. Write a C program to print sum of all odd numbers between 1 to n.
7. Write a C program to print table of any number.
8. Write a C program to find first and last digit of any number.
9. Write a C program to count number of digits in any number.
10. Write a C program to calculate sum of digits of any number.
11. Write a C program to calculate product of digits of any number.
12. Write a C program to swap first and last digits of any number.
13. Write a C program to enter any number and print its reverse.
14. Write a C program to enter any number and check whether the number is palindrome or not.
15. Write a C program to find frequency of each digit in a given integer.
16. Write a C program to enter any number and print it in words.
17. Write a C program to print all ASCII character with their values.
18. Write a C program to enter any number and print all factors of the number.
19. Write a C program to enter any number and calculate its factorial.
20. Write a C program to find HCF (GCD) of two numbers.
21. Write a C program to find LCM of two numbers.
22. Write a C program to check whether a number is Prime number or not.
23. Write a C program to check whether a number is Armstrong number or not.
24. Write a C program to check whether a number is Perfect number or not.
25. Write a C program to check whether a number is Strong number or not.
26. Write a C program to print Fibonacci series up to n terms.

8. Arrays

8.1) Write a C program to search an element in the given array (Linear Search).

8.2) Write a C program to perform matrix addition.

8.3) Write a C program to perform matrix multiplication.

Practice Programs:

1. Write a C program to read and print elements of array.
2. Write a C program to find sum of all array elements. - using recursion.
3. Write a C program to find maximum and minimum element in an array. - using recursion.
4. Write a C program to find second largest element in an array.
5. Write a C program to copy all elements from an array to another array.
6. Write a C program to insert an element in an array.
7. Write a C program to delete an element from an array at specified position.
8. Write a C program to print all unique elements in the array.
9. Write a C program to print all negative elements in an array.
10. Write a C program to count total number of even and odd elements in an array.
11. Write a C program to count total number of negative elements in an array.

12. Write a C program to count total number of duplicate elements in an array.
13. Write a C program to delete all duplicate elements from an array.
14. Write a C program to count frequency of each element in an array.
15. Write a C program to merge two array to third array.
16. Write a C program to find reverse of an array.
17. Write a C program to convert lowercase string to uppercase.
18. Write a C program to convert uppercase string to lowercase.
19. Write a C program to toggle case of each character of a string.
20. Write a C program to find total number of alphabets, digits or special character in a string.

9. Pointers

9.1) Write a C Program to Perform Addition, Subtraction, Multiplication and Division of two numbers using Command line arguments.

9.2) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.

9.3) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

10. Functions, Array & Pointers

10.1) Write a C Program to demonstrate parameter passing in Functions.

10.2) Write a C Program to find Fibonacci, Factorial of a number with Recursion and without recursion.

10.3) Write a C Program to find the sum of given numbers with arrays and pointers.

Practice Programs:

1. Program to change the value of constant integer using pointers.
2. Program to print a string using pointer.
3. Program to count vowels and consonants in a string using pointer.
4. Program to read array elements and print with addresses.

11. Strings

11.1) Implementation of string manipulation operations with library function:

a) copy
b) concatenate
c) length
d) compare

11.2) Implementation of string manipulation operations without library function:

a) copy
b) concatenate
c) length
d) compare

11.3) Verify whether the given string is a palindrome or not.

12. Structures

12.1) Write a C Program to Store Information of a book Using Structure.

12.2) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function.

13. Files

13.1) Write a C program to open a file and to print the contents of the file on screen.

13.2) Write a C program to copy content of one file to another file.

13.3) Write a C program to merge two files and store content in another file.

14. Application

Creating structures to capture the student's details save them in file in proper record format, search and prints the student details requested by the user.

Note: Draw the flowcharts using Raptor from Experiment 3 to Experiment 6.

Reference Books:

1. Let Us C Yashwanth Kanetkar, 16th edition, BPB Publications.
2. Programming in C A-Practial Approach Ajay Mittal. Pearson Education.
3. The C programming Language, Dennis Richie and Brian Kernighan, Pearson Education.
4. Problem solving using C , K Venugopal,3rd Edition,TMG Publication.

Web Links:

1. https://www.hackerrank.com/
2. https://www.codechef.com/
3. https://www.topcoder.com/
4. https://code-cracker.github.io/
5. https://raptor.martincarlisle.com/
6. https://nptel.ac.in/courses/106105085/2